Exp: 7 **Flow Control using the Sliding Window Protocol**

Aim:

To write a program to implement flow control at the data link layer using the sliding window protocol and to simulate the flow of frames from one node to another.

Requirements:

* Python 3 (or any C / Java compiler)

Algorithm:

1. Read the total number of frames and the window size.
2. Sender transmits all frames inside the current window.
3. Receiver acknowledges frames in order; the window slides forward for every ACK received.
4. If a frame or ACK is lost, the sender times out and retransmits from the lost frame (Go-Back-N).
5. Repeat until all frames are sent and acknowledged.

Program:

```
# Sliding Window (Go-Back-N) flow control simulation
import random
total  = int(input("Total number of frames : "))
window = int(input("Window size            : "))
random.seed()
base = 0
while base < total:
    end = min(base + window, total)
    print("\nSender window: frames", list(range(base, end)))
    for f in range(base, end):
        print(f"  Frame {f} sent")
    ack = base
    for f in range(base, end):
        if random.random() < 0.2:                    # 20 % chance frame/ACK lost
```

```
            print(f" Frame {f} LOST -> timeout, Go-Back to frame {f}")
            break
        print(f" ACK {f} received")
        ack = f + 1
    base = ack
print("\nAll frames transmitted successfully")
```

Result:

Flow control using the sliding window protocol was simulated successfully.